DARKSKY NEXUS

w035

Full User Manual

Comprehensive reference for every tab, decoder, and setting

1. Overview

DARKSKY NEXUS is a WebSocket bridge and signal intelligence companion for SDRplay receivers running SDRConnect, and for RTL-SDR dongles via rtl_tcp. It's distributed as a ready-to-run app (a .dmg on macOS, an installer on Windows – see the Quick Start guide) – install it like any other app, and it runs a local server in the background and presents its interface in a browser window.

Key capabilities:

- Real-time spectrum and waterfall display mirroring SDRConnect
- Full VFO / LO control – tune by clicking, typing, or using band buttons
- Signal intelligence: EIBI shortwave database, AOKI, SigIDWiki, DX clusters, space weather
- Decoder suite: CW, RTTY, PSK31, NAVTEX, FT8, WSPR, ADS-B, ACARS, AIS, HFDL, VDL2, POCSAG, Digital Voice
- fldigi integration: embedded waterfall and controls for CW/RTTY/PSK31/NAVTEX
- NEXUS / FLDIGI engine selector per decoder
- SSH Launcher: start a remote SDRConnect server over SSH
- Bookmarks: persistent named frequency bookmarks
- WSPR decoder with propagation map
- Antenna port selector for multi-antenna SDRplay devices (nRSP-ST/RSPdx)
- Tabbed category grid (COMPACT / FULL IQ / EXTERNAL) in the DECODERS panel
- PSK Reporter spot upload – FT8, WSPR, and CW decodes can be reported automatically to pskreporter.info (Section 6.6a)

NEXUS does not replace SDRConnect – it runs alongside it and communicates via WebSocket.

> **w031 – Live decoder-testing pass and PSK Reporter spot upload:** A dedicated live-testing pass exercised the Numbers-Station/HF-Intel (Rivet), FreeDV, WSPR Beacons, CW Skimmer, and Marine/VHF (AIS) decoders against real signals in Chrome, uncovering and fixing five real bugs – an unfiltered WSPR decimation step, the WSPR RTL-SDR path using the wrong sample rate, a Multimon-ng code path incorrectly running a raw-IQ discriminator on already-demodulated audio, AIS silently doing nothing in Compact/IQ-Lite mode with no warning, and a CW Skimmer peak-width filter that never scaled with zoom level and so rejected every genuine narrow CW signal. See the Troubleshooting guide for each fix in detail. This build also adds PSK Reporter spot upload (Section 6.6a) and a round of UI polish: the tab bar now resizes its headings to fill available width, the space-weather strip spaces its groups evenly instead of clustering in one corner, the AIRBAND tab's columns use proportional widths and a denser grid for utility frequencies, a dead UTC-clock element was removed from the tab bar, and yellow text in light theme now has adequate contrast.

> **w035 – New DAB/DAB+ engine, live station metadata, and two real bugs fixed**: the DAB/DAB+ decoder was rebuilt on dab_radio_nexus (a new headless tool on top of williamyang98/DAB-Radio), replacing dab-cmdline so DAB now works over a networked nRSP-ST or RTL-SDR, not just a direct RSPdx. Adds a Band III quick-tune grid with automatic background channel scanning, and a dedicated player column showing live Dynamic Label (DLS) text and MOT Slideshow station-logo/album-art images. Also fixes two bugs found during live testing: classic DAB (MP2) stations were silent while DAB+ played fine, and the scanner could misattribute a previously-locked ensemble's data to the wrong channel. See Section 6.17 for details. Also adds HD Radio (NRSC-5) hybrid digital AM/FM decoding via nrsc5_nexus (built on theori-io/nrsc5) with simultaneous multi-program (HD1–HD8) audio and metadata – see Section 6.20. Also adds DRM/DRM+ decoding via dream_nexus (Section 6.19). As of 2026-08-03, packaged macOS releases bundle all three

companion engines (DAB, HD Radio, DRM) rebuilt with a pinned deployment target, so they launch correctly on older macOS versions (Sequoia, Sonoma), not just the build machine's own.

w0.2.3 – IQ Lite confirmed non-functional by design, Full IQ now default: Earlier builds assumed SDRConnect's "IQ Lite" stream mode delivered a lower-bandwidth raw IQ stream. Direct testing against SDRplay's own reference client, packet capture, and direct confirmation from SDRplay support all agree: **IQ Lite and Compact modes never deliver raw IQ (type-2) frames on the nRSP-ST** – they are demodulated/decoder-rate audio-only modes. This was previously misattributed to the same issue as SDRplay support ticket #726970; it is not the same issue, it is expected behaviour for those stream modes. **Full IQ mode works correctly** – direct testing sustained 2 MSps over Wi-Fi with zero dropped frames – and is now the default stream mode NEXUS selects for SDRplay devices. See the Troubleshooting guide for the full history and current status of ticket #726970.

2. Architecture

This section describes what happens under the hood — useful for troubleshooting, not required reading to use NEXUS. Whether you installed the .dmg/.exe or are running the Python source directly, the internals are the same; the diagrams below refer to w035_NEXUS.py, which is what's bundled inside the installed app.

```
SDRplay RSPdx
      │
SDRConnect (port 5454 WebSocket)
      │
w035_NEXUS.py ←──────────────────── Your browser (port 8888/8889)
      │                            │
      ├── HTTP server (port 8888) ──────▶ serves DARKSKY_NEXUS.html
      ├── WebSocket bridge (port 8889) ─▶ bi-directional UI control
      ├── rigctld server (port 4532) ───▶ WSJT-X / fldigi CAT polling
      └── fldigi XML-RPC (port 7362) ───▶ fldigi control (outbound)
```

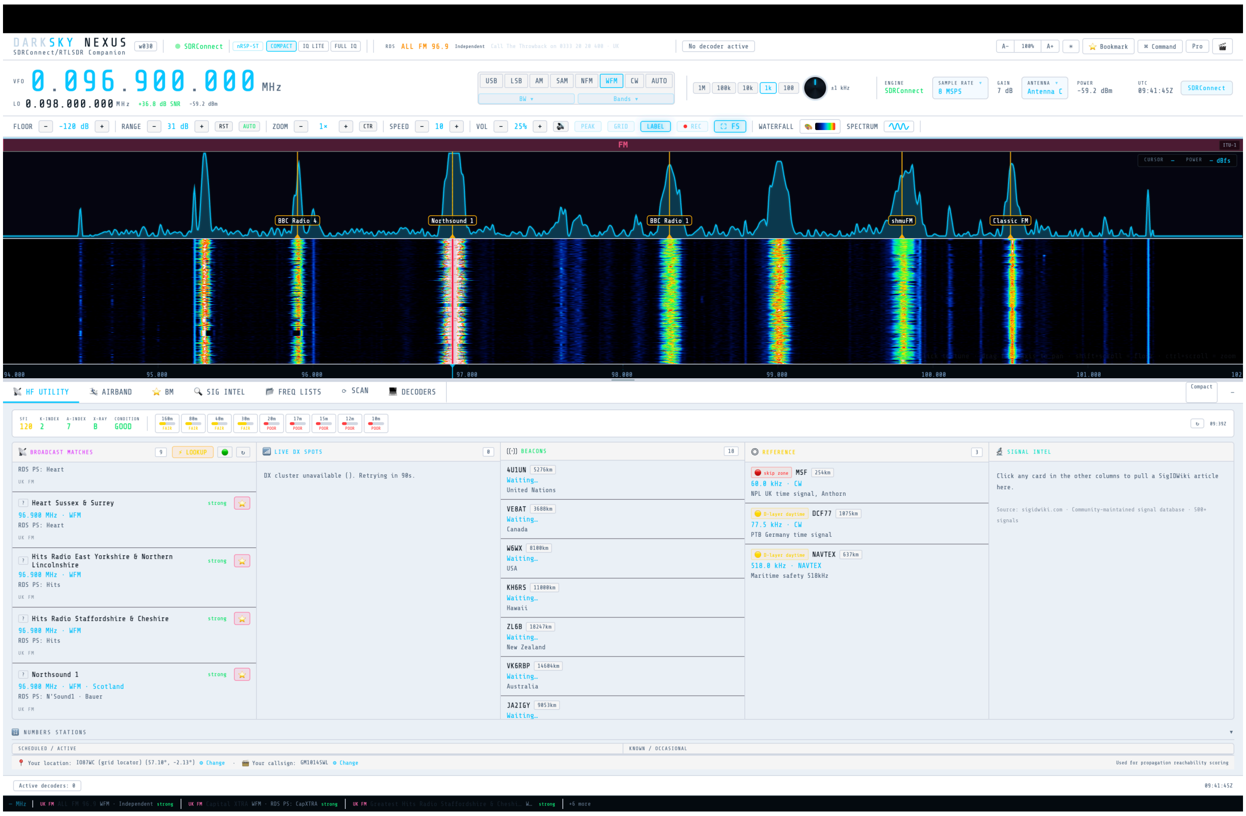
For RTL-SDR:

```
RTL-SDR dongle
      │
rtl_tcp (port 1234)
      │
w035_NEXUS.py  (--rtl flag)
```

Data files:

- **Installed app (.dmg/.exe):** managed automatically — no need to look for these. They live in ~/Library/Application Support/DARKSKY NEXUS/ on macOS, or %APPDATA%\DARKSKY NEXUS\ on Windows.
- **Running from Python source:** the same files live in the same folder as w035_NEXUS.py instead.
- eibi.csv — EIBI shortwave schedule (auto-downloaded)
- airports.csv — OurAirports database (auto-downloaded on first AIRBAND use)
- airport-frequencies.csv — Airport frequency data
- darksky_bookmarks.json — Your saved bookmarks
- darksky_config.json — Persisted settings

3. Main Interface



Main interface, full view – HF UTILITY tab, waterfall, and status bar

3.1 Top Bar

Left side:

- DARKSKY NEXUS logo and version badge
- Connection status indicator (SDRConnect / RTL-SDR / OFFLINE)
- Stream mode indicator (Compact / Full IQ)
- Active decoder badge

Right side:

- **UI size control (A– / percentage / A+)** – scales the entire app interface up or down, from 80% to 160%. Added in w0.2.6 in response to beta feedback that text was too small even on a 27" 1920×1080 monitor. Your chosen size is remembered across launches. Unlike the waterfall zoom controls (section 3.4), this resizes every panel, button, and label in the app – it's a whole-UI scale, not a frequency-span zoom.
- 🌙 **Theme toggle** – switches between dark and light UI themes. Persisted across launches. Yellow status/warning text (used across several tabs) had poor contrast against the light theme's background – fixed in w031 with a darker yellow specific to light theme.
- ⭐ **Bookmark** – saves the currently tuned frequency/mode as a bookmark in one click (see Section 8, Bookmarks).

- ⌘ **Command** – opens the Command Palette (also **Ctrl/⌘+K**): a searchable list of app-wide actions (tune to frequency, toggle Pro Mode, toggle Cinematic Mode, switch tabs, and more) without hunting through menus. See Section 14 for the full shortcut list.
- **Pro** – toggles Pro Mode: an extra DSP-controls strip (Noise Reduction, Squelch, AGC, low-cut filter, NFM de-emphasis, WFM stereo, audio limiter) below the mode buttons, plus a Pro-only DSP Graph view showing the live audio-processing chain. See Section 3.4a.
- 🎬 **Cinematic Mode** (also keyboard **C**) – a full-screen ambient visualisation: the waterfall/spectrum take over the whole window with the frequency, mode, sample rate, SNR and engine shown as an overlay, tab/panel chrome hidden. Five visual scenes are available (NEXUS, Retro, Bars, Phosphor, Polar) – cycle with number keys **1-5** while active, or click anywhere to exit.
- UTC clock
- SDRConnect label

3.1a Status Bar – Sample Rate & Antenna (SDRplay devices)

Above the waterfall, the status bar (ENGINE / SAMPLE RATE / GAIN / ANTENNA / POWER / UTC) includes two controls specific to SDRplay devices connected via SDRConnect:

SAMPLE RATE – click to open a dropdown of available rates. In **Full IQ** or **Compact** mode, this sets the actual hardware sample rate. (Prior to w0.2.3, an "IQ Lite" mode existed with a fixed 192 kSPS hardware rate and a display-span picker in its place; IQ Lite has been confirmed to never deliver raw IQ on the nRSP-ST and Full IQ is now the default – see section 1.)

ANTENNA – click to open a dropdown of antenna ports (e.g. Antenna A, Antenna B, Antenna C on nRSP-ST/RSPdx). Selecting one switches the active RF input on the device. This control is hidden automatically for devices that don't report multiple antennas (e.g. RSP1A) – it only appears once SDRConnect reports a `valid_antennas` list for the connected device.

3.1b RDS Decode (FM Broadcast)

When tuned to a WFM station carrying RDS (Radio Data System), a compact RDS strip appears inline in the top bar, showing:

- **PS** – the station's 8-character Program Service name (e.g. "BBC R4")
- **PTY** – the Program Type code (e.g. News, Pop Music), decoded to a readable label
- Radiotext – the scrolling free-text message some stations send (song title/artist, slogans, etc.)

The strip only appears in **WFM** mode and only once RDS data is actually present – it stays hidden otherwise, so it never shows stale or guessed information as if it were RDS.

RDS also feeds two other places automatically:

- **HF UTILITY → BROADCAST MATCHES** column – once RDS locks, the received PS name is authoritative there. Before lock (or for stations that don't carry RDS), NEXUS instead shows a likely station name from a small built-in UK FM frequency/PI database, so you still get a probable identification while waiting for a real RDS decode.
- **Auto-bookmark** – a station identified via RDS (or the fallback UK FM database) can be automatically suggested as a bookmark, carrying the decoded name rather than a bare frequency.

3.2 VFO / Frequency Display

The large cyan digit display shows the current VFO (tuned signal) frequency in MHz.

Below it, the LO line shows the hardware centre frequency and SNR/signal power.

To change frequency:

- Click the display and type a frequency in MHz, then press Enter
- Click anywhere on the waterfall
- Use band buttons in the toolbar

3.3 Mode & BW Controls

The mode button group (USB / LSB / AM / SAM / NFM / WFM / CW / AUTO) sets the demodulation mode. Click a mode to activate it.

BW button opens the bandwidth preset panel – select a preset or type a custom value.

Bands button opens the band plan panel – click a band to jump to its centre frequency.

3.4 Toolbar

Left group:

- FLOOR (±) – noise floor in dBfs. Lower reveals weaker signals; too low shows noise artefacts.
- RST – reset floor to -120 dBfs
- AUTO – auto-set floor from current noise average
- RANGE (±) – dynamic range in dB. 20-40 dB suits most signals.
- ZOOM (±) – spectral zoom factor (1× to 8×)
- CTR – centers the visible waterfall/spectrum span on the currently tuned VFO frequency, at any zoom level (including 1×). Added in w0.2.6 in response to beta feedback that the tuned signal was always stuck at the left or right edge instead of centered. This only changes what's displayed – it does not retune the radio or move the hardware centre frequency. Previously, recentring on the VFO only happened automatically as a side effect of zooming in past 1×; CTR makes that available on demand at any time. Zooming back out to 1× (or pressing RST on zoom) releases the centring and returns to the default hardware-centred view.
- SPEED (±) – waterfall scroll speed
- VOL (±) – audio volume (RTL-SDR path only)

Right group:

- PEAK – enable peak hold on spectrum trace
- GRID – frequency grid overlay
- LABEL – signal labels on waterfall
- REC – record to a file. See 3.4a1 below for the AUD/IQ mode pill next to it.
- FS – fullscreen
- WATERFALL – toggle waterfall display
- Colour swatch – click to open the waterfall colour palette picker. Eight palettes are built in: Classic SDR (the default – blue/green/yellow/red), Heat, Viridis, Greyscale, Midnight Blue, Solar, Inferno, and DARKSKY Neon. Your choice is remembered across sessions and applies to both the waterfall and the CW decoder's own separate mini-waterfall.
- SPECTRUM – toggle spectrum trace
- Waveform icon – toggle oscilloscope view

3.4a1 Recording (REC): AUD / IQ modes (w032)

A small **AUD | IQ** pill sits next to the REC button – pick a mode, then REC starts/stops a recording in that mode. The choice is remembered between sessions.

- **AUD** – saves the demodulated audio for whatever you're currently listening to, as a WAV file. Works in any stream mode (Compact, IQ Lite, or Full IQ).
- **IQ** – saves the raw, unfiltered Full IQ stream to a `.cf32` file (interleaved 32-bit float I/Q samples, no header) – the same samples SDRConnect delivers, before any of NEXUS's own filtering. Requires Full IQ device mode; REC will show an error toast instead of starting if you're in Compact or IQ Lite.
- Files save to a `recordings` folder next to the NEXUS script, named `darksky_rec_<mode>_<timestamp>.wav` or `.cf32`. Stopping the recording shows a toast with the saved path, duration, and file size.

> IQ mode exists mainly so you can feed a capture into external SDR software (e.g. AIS-catcher, GNU Radio) for comparison against NEXUS's own decoders, without both programs needing simultaneous exclusive access to the same SDR hardware. AIS-catcher reads it directly with `-r -ga FORMAT CF32 FILE <path>` plus `-s <sample_rate>` set to whatever rate the saved-file toast reported.

3.4a Pro Mode

Clicking **Pro** in the top bar (Section 3.1) reveals an extra DSP-controls strip between the mode buttons and the band plan strip:

- **NR** – noise reduction on/off, with a strength slider (0-100)
- **SQL** – squelch on/off, with a threshold slider in dBm (−120 to 0)
- **AGC** – automatic gain control on/off
- **LOW-CUT** – high-pass filter cutoff in Hz, adjustable in 50 Hz steps
- **DE-EMPH** – NFM de-emphasis filter toggle
- **STEREO** – WFM stereo decode toggle
- **LIMIT** – audio limiter toggle, to prevent clipping on strong signals

All of these send live settings to the backend as you change them – there's no separate "Apply" step. A companion **DSP Graph** view (opened from the same Pro Mode area) shows the current audio-processing chain visually, stage by stage, reflecting each toggle's on/off state live as you change it – useful for seeing at a glance which stages are actually active in the current audio path.

3.4b Band Plan Strip

The thin coloured strip directly above the spectrum/waterfall shows the amateur/broadcast band plan for your selected ITU region – hover a segment for the band name, click to tune there. A small pill in its top-right corner (**ITU-1 / ITU-2 / ITU-3**) shows the active region; click the pill to cycle regions if you operate somewhere other than the default.

3.5 Waterfall & Spectrum

The waterfall scrolls downward in real time. The spectrum trace runs across the top.

Mouse interactions:

- **Left-click** – tune VFO to clicked frequency
- **Right-click** – save bookmark at clicked frequency
- **Scroll** – scroll speed (default); **Shift+Scroll** = floor; **Ctrl+Scroll** = zoom
- **Drag on frequency axis** – pan the display left/right
- **Drag resize handle** (between waterfall and panel area) – resize waterfall height. Height is remembered across sessions.

The tune line (red vertical) shows the current VFO. The BW overlay shows the current bandwidth.

3.6 Frequency Axis

The grey axis below the waterfall shows frequency labels. Drag left/right to pan; click to tune.

3.7 Tab Area

Below the waterfall: HF UTILITY | AIRBAND | BM | 📂 FREQ LISTS | ⟳ SCAN | SIG INTEL | DECODERS

The spectrum and waterfall sit above this tab row and stay visible no matter which tab is active – switching tabs only changes the panel below, never the spectrum display.

Tab headings fill the available width (w031): tab labels now grow evenly to fill the full width of the tab bar and reflow automatically as the window is resized, rather than staying left-clustered with empty space to the right.

4. HF Utility Tab

The default tab when NEXUS opens. Above the five-column grid, a status strip shows space weather (SFI, K-index, A-index, X-ray, overall CONDITION) and a per-band propagation matrix, with a ↻ refresh button (fetches latest NOAA SWPC data) and a last-updated time. As of w031 this strip spaces its groups (stats, propagation pills, status readout) evenly across the full width instead of clustering them in one corner.

Below that, a five-column grid:

Column 1 – Broadcast Matches (⚡ LOOKUP / filter / refresh buttons in its header)

- Shows EIBI/AOKI shortwave schedule entries and RDS-identified FM stations near your tuned frequency
- Once RDS locks on a WFM station (Section 3.1b), the live-decoded PS name is shown as authoritative; before lock, a likely name from the built-in UK FM database is shown instead, tagged so it's clear it isn't a confirmed RDS decode yet
- ⚡ **LOOKUP** queries the EIBI+signal database for the currently tuned frequency; the 🟢 filter button restricts results to "likely receivable" only; ↻ re-downloads the EIBI/AOKI database from eibispace.de

Column 2 – Live DX Spots

- Live spots from the DX cluster network; click a spot to tune to it

Column 3 – Beacons

- The NCDXF 18-beacon world schedule; click a beacon to tune to it

Column 4 – Reference

- Nearby reference/time-standard channels for your tuned frequency (e.g. MSF, DCF77, NAVTEX) with distance and a skip-zone/daytime propagation hint

Column 5 – Signal Intel

- Click any card in the other four columns to pull its SigIDWiki article here – signal characteristics, sample images, modulation notes (source: sigidwiki.com, 500+ signal community database)

Live Decode (appears below the grid only once a decoder is active)

- Mirrors decoded text from whichever decoder is currently running, with a signal-class badge, a ⚡ **ID Signal** button (same EIBI/signal lookup as Column 1, for the tuned frequency), and a **Clear** button

Numbers Stations (collapsible section below the grid, click the header to expand/collapse)

- Two panels: **Scheduled / Active** and **Known / Occasional** – a reference list of numbers-station schedules, separate from the live decoder in Section 6.14 (Numbers-Station/HF-Intel)

Location & callsign bar (bottom of the tab)

- 📍 **Your location** – click ⚙️ **Change** to set your location by city name, Maidenhead grid locator (e.g. `IO87wc`), or raw `lat,lon`. Used to score propagation reachability by band throughout the HF Utility tab (Column: Propagation matrix) and to centre distance/bearing calculations elsewhere in the app.
- 📻 **Your callsign** – click ⚙️ **Change** to set your own callsign (validated as 3–12 characters, letters/digits/`/` only). Saved locally in your browser, and – as of w031 – also used as the sending callsign for PSK Reporter spot uploads if reporting is enabled (Section 6.6a).

Custom frequency lists were originally a collapsible section here (added in w0.2.1). As of this build they have moved to their own dedicated tab – see Section 9, Frequency Lists Tab – with a searchable/filterable table, click-to-tune, and location-aware region tagging. Imported-list entries still surface in SIG INTEL / SW SCHEDULE lookups alongside EIBI/SigID results, tagged with source CUSTOM, exactly as before.

5. Signal Intelligence Tab

Three-column layout:

Left – Database Results

- EIBI shortwave schedule entries near the tuned frequency (±5 kHz window)
- AOKI entries
- BUILTIN_SIGNALS (utility/military/maritime/aviation/digital)
- Click any entry to tune

Centre – SigIDWiki

- When a signal is identified, fetches the SigIDWiki article inline
- Shows signal characteristics, samples, modulation

Right – Active Signals

- Signals currently visible in the spectrum above noise threshold
- Auto-updated from waterfall data

6. Decoders

Access via the **DECODERS** control in the tab bar. Each decoder activates a dedicated panel.

> **w033 – DECODERS panel redesigned:** clicking **DECODERS** now opens a floating panel with category tabs (COMPACT / FULL IQ / EXTERNAL) across the top and a grid of decoder buttons below – the same interaction as the **BANDS** control, so picking a decoder now works exactly like picking a band. This replaces w032's dropdown-with-collapsible-headers design; the underlying category groupings and which decoders live in each are unchanged (see below), only the way you get to them. w032 keeps the original dropdown-menu style – the panel redesign is w033-only. Opening either DECODERS or BANDS automatically closes the other if it's open.

Decoders are grouped into three categories:

- **COMPACT** – audio-only decoders that work even when SDRConnect is in Compact mode (no IQ stream required): FT8/WSPR, WSPR Beacons, CW, RTTY, PSK31/PSK63/PSK125, NAVTEX, Olivia/Contestia/MFSK/Hell/DominoEX, Marine/VHF, Multimon, Numbers-Station/HF-Intel (Rivet), FreeDV.
- **FULL IQ** – decoders that require a genuine raw IQ stream and will not work in Compact mode: WEFAX, SSTV, ACARS, Pager/POCSAG, AIS. (Prior to w0.2.3, WEFAX was listed under a separate "IQ LITE+" category, on the assumption that IQ Lite mode could supply it with a usable IQ stream. That category has been removed – IQ Lite never delivers raw IQ on the nRSP-ST, so WEFAX now correctly sits alongside the other genuinely IQ-dependent decoders here, all of which need Full IQ mode.)
- **EXTERNAL** – decoders reading a separate feed/process, not gated by stream mode: ADS-B, DAB/DAB+, DRM/DRM+, HD Radio (NRSC-5), HFDL, VDL2, ISM Sensors (rtl_433), Digital Voice (DSD+), Trunked P25/DMR/NXDN.

6.1 Engine Selector

Decoders that support fldigi (CW, RTTY, PSK31, NAVTEX) show a **NEXUS | FLDIGI** toggle in their header.

- **NEXUS** – uses NEXUS internal decoder. No external software needed. Shows NEXUS-specific controls (tone scope, parameters, etc.).
- **FLDIGI** – routes audio through fldigi via XML-RPC. Shows embedded fldigi waterfall and controls. Requires fldigi running with virtual audio routing.

The selected engine is saved per-decoder and restored on next launch.

Gotcha: because the engine choice persists across sessions, it's easy to leave a decoder on FLDIGI without realizing it – there's nothing on screen pointing this out until you press Start. If CW or RTTY is on FLDIGI, pressing Start routes through fldigi instead of the NEXUS decoder: no NEXUS waterfall, no NEXUS decoded-text panel, no error. As of this build, NEXUS shows a toast warning when this happens ("CW engine is set to fldigi – switch to NEXUS for the built-in waterfall/Skimmer"); switch the toggle to NEXUS and press Start again. See the Troubleshooting guide's "No decoded text from CW / RTTY" entry, Cause 0.

6.2 CW Decoder

NEXUS engine:

- Tone frequency slider (300–1200 Hz) – set to match CW tone pitch
- Key threshold slider – sensitivity (2–20 dB)
- Speed trend sparkline (WPM history)
- Last chars display (large amber text)
- Activity canvas
- Sweep mode: steps VFO through a frequency range looking for CW activity

Skimmer peak-width filter fixed (w031): the CW Skimmer's candidate-peak detector rejects anything wider than a hard-coded width threshold, on the assumption that a genuine CW carrier is narrow and interference/noise peaks are wide. That threshold didn't scale with the waterfall's actual frequency resolution (Hz per FFT bin), which varies with zoom/span – at typical zoom levels a single bin alone could exceed the threshold, so even a strong, genuinely narrow CW signal was always classified as "too wide" and silently discarded, leaving the Skimmer channel count stuck at zero even with clear activity on screen. The width threshold now scales with the current Hz-per-bin resolution, so real narrow CW carriers are correctly recognised at any zoom level.

Mini-waterfall freeze hardening (w0.2.6): some users saw the CW mini-waterfall freeze permanently right after pressing Start. No single reproducible cause was confirmed, but the most likely failure mode – a one-time rendering exception leaving shared waterfall state (the colour-map lookup table) stuck in a bad state, with every later frame silently failing the same way – is now guarded against: rendering errors are caught and logged to the browser console instead of failing silently, and a failed colour-map rebuild falls back to a built-in per-pixel colour function rather than leaving the waterfall stuck. If you ever see this again, check the browser console for a logged warning and report it.

Mini-waterfall + spectrum: fixed ±3000 Hz close-up scope, immune to the BW setting (w031): the CW mini-waterfall, and a new line-trace spectrum display above it, are both driven by a server-computed FFT taken from the genuine, unfiltered receiver IQ signal – the same signal the CW decoder mixes down for its own decode filter, captured before that filter narrows it. On IQ-Lite and Full IQ streams this gives a fixed ±3000 Hz view centred on the VFO that doesn't change as the CW BW setting is narrowed or widened, mirroring FT8 INTERNAL's own waterfall. (On Compact-mode streams, no unfiltered IQ is available – SDRConnect has already demodulated to audio by the time NEXUS sees it – so the view there is built from that demodulated audio instead, and does narrow along with the BW setting; this is a stream-mode limitation, not a bug.) The Compact-mode broadcast was also fixed in this build to always run once decode starts, so on every stream mode the mini-waterfall is guaranteed to have data as soon as CW decode is active – it no longer depends on FT8 INTERNAL's own audio feed being on.

Keying-scope layout no longer jumps (w031): the TONE/SPEED/SNR stat boxes above the keying scope now have a fixed minimum width and tabular (fixed-width) digits, so the scope no longer shifts sideways each time a readout gains or loses a digit.

CW tab layout – 3 sub-columns + SKIMMER/SINGLE toggle (w031): the NEXUS-engine panel is now 3 side-by-side sub-columns – waterfall + zoom/floor/colour controls | tone/threshold + Last Chars + sweep config | speed trend + Skimmer Channels – so nothing stacks below the mini-waterfall and no scrolling is needed to reach the controls. The single-channel decoder's own decoded-text panel is no longer permanently on screen; a **SKIMMER / SINGLE** toggle in the third sub-column switches between the multi-channel Skimmer Channels list (default) and the single-channel decoder's own text.

Selecting the FLDIGI engine always shows the single-channel text regardless of the toggle, since fldigi has no Skimmer pool of its own. The toggle choice persists across reloads.

Skimmer candidate detection now uses real SNR (w032): the Skimmer's candidate-peak scan previously ran against the mini-waterfall's own display bins – a self-normalising, log-compressed array with no fixed relationship to actual signal-to-noise ratio, so most promoted candidates measured only 0–1.5 dB real SNR against the decoder's 8 dB decode threshold and sat in the channel list indefinitely without ever producing text. On IQ-Lite/nRSP-ST connections, candidate detection now runs server-side against the same real-dB wideband FFT array the mini-waterfall's own data comes from, before that data is compressed for display – so a candidate's SNR reading is the same number the decoder itself will judge it against. Full IQ and Compact-mode streams still use the previous display-bin scan as a fallback; this is a scoping decision, not an oversight, and may be extended to those modes in a future build.

Unmatched Morse patterns no longer silently dropped (w032): a timing error – a QSB-distorted dash, keyer weighting, a dropped dit – that produced a symbol pattern not matching any character used to render nothing at all, indistinguishable on screen from a normal gap between words. It now renders # instead, so a run of timing errors shows up as a run of #s rather than an invisible gap. A genuinely empty gap (nothing sent) still stays silent.

Persistent "Tuned Decode" ticker (w032): a single-line, tail-truncated readout of the single-channel decoder's live text now sits between the CW stats bar and the three sub-columns, visible regardless of which SKIMMER/SINGLE view is selected – previously the only way to see the dial-frequency decode was to switch to SINGLE view, which hid the Skimmer channel list at the same time.

Skimmer channel markers on the mini-waterfall (w032): each active Skimmer channel now shows a coloured tick and a callsign/SNR label (or its dial frequency, before a callsign is extracted) directly on the mini-waterfall, positioned at its real frequency offset from the VFO. The colour matches that channel's row in the Skimmer Channels list, so a signal on the waterfall and its decoded text in the list are visibly the same channel at a glance – only shown in SKIMMER view.

Waterfall toolbar consolidated, TONE demoted, scan range surfaced, waterfall enlarged (w032): the zoom/colour and floor/range controls, previously two stacked full-width rows, are now one row with a divider between the two groups. The read-only TONE readout shrank to a small inline line so it no longer competes visually with the Key Threshold slider beneath it. The Skimmer scan-range readout moved from a small muted line inside the Skimmer Channels panel (which disappears in SINGLE view) to a coloured chip next to the SKIMMER/SINGLE toggle in the always-visible tab header. The space freed by consolidating the toolbar went into the mini-waterfall (220px → 260px) and its spectrum trace (70px → 90px).

Mini-waterfall resolution fixed (w032): the server-computed FFT behind the mini-waterfall (and RTTY tone scope, and fldigi waterfall) natively resolves to roughly 512 bins at ~11.7 Hz/bin on IQ-Lite/Full IQ connections, but was being resampled down to a fixed 256 bins before being sent to the browser – discarding about half the real resolution, which the browser canvas then interpolated back up to display width anyway. That resample step is removed; the mini-waterfall, RTTY tone scope, and fldigi waterfall now all show genuine full-resolution data.

Mini-waterfall zoom is now a real "zoom FFT" (w032): the −/+ zoom control above the mini-waterfall used to be purely cosmetic – the backend always sent the same fixed-resolution bins and zooming just cropped/interpolated that same data client-side. It's now a genuine zoom: the backend scales its FFT size up with the zoom level (base 4096-pt at 1×, up to 32768-pt at 8×, on IQ-Lite/Full IQ connections) and runs it over a longer window of raw IQ, giving real additional detail as you zoom in rather than a blurrier crop of the same 512 bins. This is a genuine trade-off, not a free upgrade: a longer window needed for finer frequency resolution is unavoidably a longer window in time too (the same trade-off any SDR panadapter's zoom makes), so the mini-waterfall updates more slowly and blurs fast keying more at high zoom – this is expected, not a bug, and the mini-waterfall's job is identifying where a signal sits, not showing individual key-down timing.

FLDIGI engine:

- Embedded fldigi waterfall (160px, 5fps live)
- WPM slider (5–60)
- AFC and squelch toggles
- Squelch level slider
- Decoded text output

Start/Stop button in header activates the selected engine.

6.3 RTTY / FSK Decoder

NEXUS engine (left + centre columns):

Parameters (left column):

- Baud rate (default 45.45, common: 50, 75, 100, 110)
- Shift in Hz (default 170, common: 200, 425, 850)
- Mark frequency in Hz (default 2125)
- Auto-detect baud & shift checkbox
- Charset: ITA-2, US-TTY, MTK-2, ASCII, SITOR-B, NAVTEX
- Known Signals preset list

Tone Scope (centre column):

- Waterfall/spectrum display showing audio spectrum
- Toggle WATERFALL / SPECTRUM mode
- Mark (M) and Space (S) draggable markers – drag to set tone frequencies
- Zoom buttons (−/+) – 5 levels: 500Hz, 1kHz, 2kHz, 3kHz, 4kHz span
- Confidence bar
- **Signal: LOCKED / NO SIGNAL badge + dB readout (added July 2026)** – RTTY has a continuous carrier (mark or space is always being sent), unlike CW's genuine on/off keying, so earlier builds had no way to tell noise from a real signal and would happily grind out plausible-looking garbage text from an untuned or dead frequency. A live spectral squelch now checks a third reference point in the passband that should be empty during a real signal; the badge reads **LOCKED** (green) with the current SNR in dB once that check passes, or **NO SIGNAL** (red) otherwise. New character acquisition only starts while LOCKED – an already-in-progress frame is left alone. If you see NO SIGNAL, retune or check your tone/shift settings before assuming the decoded text (if any lingers on screen) is meaningful.

FLDIGI engine (replaces centre column):

- Embedded fldigi waterfall

- Carrier and BW readouts
- AFC and squelch toggles
- Squelch level slider
- Decoded text output

6.4 PSK31 / PSK63 / PSK125

Sub-mode selector in header: PSK31 | PSK63 | PSK125.

NEXUS engine: Quick-tune frequency list, decoded text.

FLDIGI engine: Embedded fldigi waterfall, AFC/SQL controls, decoded text.

6.5 NAVTEX

Fixed parameters: 100 Bd, 170 Hz shift, USB, ~518/490/4209 kHz.

NEXUS engine: Standard frequencies list (518 kHz international, 490 kHz national, 4209.5 kHz HF).

FLDIGI engine: Embedded fldigi waterfall and controls.

6.6 FT8 / WSPR

The FT8/WSPR tab has an **FT8 SOURCE** toggle at the top: **WSJT-X** (the original bridge, described below – default) or **INTERNAL** (new in w030 – a native decoder that needs no other software at all).

FT8 – WSJT-X source (bridge mode):

- Requires WSJT-X running **on the same computer as NEXUS** (see below)
- NEXUS provides frequency control and decode display
- Grid map shows decoded stations on a Leaflet world map
- Click any spot on the map or table to tune
- **Map style dropdown** (added July 2026) above the map – same five no-API-key providers as the FT8 INTERNAL map (CARTO Dark/Light/Voyager, OpenStreetMap, Esri Satellite), chosen independently of that map's own style setting.

How NEXUS and WSJT-X talk to each other (two separate, one-way channels):

NEXUS does not have a single combined link with WSJT-X – frequency control and decode display use entirely different mechanisms, and either can work while the other doesn't:

- **NEXUS → WSJT-X (frequency control):** NEXUS sends UDP Heartbeat and DialFrequency messages to `127.0.0.1:2237`, the standard WSJT-X UDP server port. This lets you tune WSJT-X from NEXUS and remotely set its mode. It is outbound only – WSJT-X does not reply over this port.
- **WSJT-X → NEXUS (decodes):** NEXUS reads decodes by tailing WSJT-X's own `ALL.TXT` log file on disk (polled every ~2 seconds for new lines), not by listening on a network port. WSJT-X writes every decode it makes to this file automatically; NEXUS just watches it.

Because the decode side reads a local file, **WSJT-X must be running on the same machine as NEXUS** – there is no remote/network equivalent for this half of the link, even though frequency control over UDP works fine to a WSJT-X on another machine. The default search locations for ALL.TXT are:

```
macOS:   ~/Library/Application Support/WSJT-X/ALL.TXT
         ~/Documents/WSJT-X/ALL.TXT
Linux:   ~/.local/share/WSJT-X/ALL.TXT
Windows: %APPDATA%\WSJT-X\ALL.TXT
```

ALL.TXT is only created by WSJT-X after its first real decode, so a freshly-started WSJT-X with nothing decoded yet will show no spots in NEXUS until that happens. File-tailing also only begins once the FT8/WSPR decoder panel is started in NEXUS – having frequency control working already does not mean the decode reader is active. See the Troubleshooting guide for the full diagnostic checklist if spots aren't appearing.

FT8 – INTERNAL source (native decoder, added in w030):

Selecting INTERNAL switches to a self-contained FT8/FT4 decoder that runs entirely inside your browser – no WSJT-X install, no ALL.TXT, no UDP port, nothing else to launch. It uses ft8ts (a from-scratch JavaScript/TypeScript port of the WSJT-X v2.7 FT8/FT4 algorithm, © Roger Need, GPL v3, https://github.com/e04/ft8ts), executed inside a dedicated Web Worker so decoding doesn't block the UI.

The audio it decodes does not come from your computer's microphone, and there is no separate audio device to select. Its origin is the same WebSocket connection NEXUS already uses for everything else (ws://127.0.0.1:8889 by default) – w035_NEXUS.py taps the same demodulated audio stream that feeds the CW/RTTY/AIS decoders and streams it to the browser as a new binary frame type (byte[0]=0x02, distinct from the existing 0x01 spectrum-bin frames), only while INTERNAL mode is actually selected and Start is pressed in the tab. Practically, this means: tune to an FT8 (or FT4) frequency, set mode to USB, switch FT8 SOURCE to INTERNAL, and click ▶️ **Start** – no other setup step exists.

- Works over a remote --sdr connection or the SSH Launcher just as well as bridge mode does for frequency control – since it doesn't depend on a local ALL.TXT file, it's actually less location-sensitive than WSJT-X-bridge mode.
- You'll hear the FT8/FT4 audio tones play through your browser while decoding, the same way WSJT-X plays audio through a soundcard.
- ▶️ **Start / ■ Stop** controls the decoder – matching every other decoder in NEXUS (CW, RTTY, etc.). This replaced a single "Decode On/Off" toggle that was hard to read in both light and dark themes.
- Decoded stations are grouped and colour-coded by DXCC entity (country), looked up from the callsign prefix – this now works correctly; a plain grid-square or country-centroid position is shown on the map either way.
- On the RTL-SDR engine specifically, decode timing/frequency can be very slightly (roughly ±1-2%) less precise than on the SDRConnect/SDRplay paths, because of how RTL-SDR's sample-rate decimation rounds – this is a minor precision note, not a decode failure.

- **MON VOL slider (added w031)** – a small volume control next to the Clear button in the toolbar sets how loud the decoded audio plays through your speakers, independently of the SDRConnect/receiver volume. It defaults to 50% and is remembered across sessions. This is purely a local playback preference – turning it down never affects what the decoder itself analyses, only what you hear.

Audio distortion fixed (w031): earlier builds could sound audibly distorted when FT8 INTERNAL decode started, especially with SDRConnect's own volume set low – a gain-compensation curve meant to keep quiet volume settings audible had no upper limit and could over-boost the signal past full scale. The boost is now capped and uses a soft-knee curve that rounds off loud peaks instead of clipping them harshly; combined with the new MON VOL slider above, this should resolve any remaining distortion without needing to lower your computer's own system volume.

Band Hopping (FT8 INTERNAL)

The **Hop Off / Hop On** button opens a Band Hopping window – an unattended multi-band scan, not a single on/off switch. Click it to configure a table of rows, each with an on/off checkbox, a **Band**, a **Mode** (FT8 or FT4), and **Cycles** (how many 15s/7.5s decode cycles to spend on that band before moving on), plus live per-row **Decodes** / **Callsigns** / **DXCC** counts.

- Click **Start** in the Band Hopping window to begin – it retunes the SDR itself for each enabled row in turn (sample rate, centre/VFO frequency, USB demod, and bandwidth all set automatically to that band's FT8/FT4 dial frequency), starts the decoder if it isn't already running, and loops back to the first row after the last.
- While hopping is active, Start/Stop, the mode toggle, and the manual band selector on the main panel all lock out – the Band Hopping window's own **Start** button (now reading **Stop**) is what controls it instead.
- Each band switch drops a separator line into the decode list rather than clearing it, so a hopping session builds up the same kind of multi-band picture manual band-switching does (Section 6.6's map notes below).
- **Reset Stats** clears every row's Decodes/Callsigns/DXCC tally without touching the row configuration itself; the 🗑 icon on a row removes that row.

Propagation Map (FT8 INTERNAL)

The map column to the right of the decode list (added July 2026) has grown a few controls of its own:

- **Drag-resize handle** – a thin vertical splitter between the decode list and the map column. Drag it to make the map wider (useful on wide displays where the decode list otherwise leaves unused space) or narrower; the chosen width is remembered across sessions.
- **Spots persist across band changes** – switching bands with the quick-tune buttons (160m–6m, below the decode list) used to wipe every marker and decoded row from the map. It no longer does: switching bands now just adds a separator line to the decode list and keeps accumulating, exactly like Hop mode already did, so a session can build up a multi-band picture – each station's marker is colour-coded by the band it was heard on. Click **Clear** to wipe everything deliberately.
- **Map style dropdown** (top of the map column) – switches the base map between five tile providers, all free and requiring no API key: CARTO Dark (the default), CARTO Light, CARTO Voyager, OpenStreetMap, and Esri Satellite imagery. Your choice is remembered across sessions. The same picker is also available on the FT8/WSPR bridge (WSJT-X source) map further up this section, chosen independently.

"My QTH" marker (added w031): every map in NEXUS – this one, the FT8/WSPR bridge map, the WSPR Beacons propagation map, and the ADS-B, AIS, ACARS, and FT8 Skimmer/station maps described later in this section – now shows a distinguishing orange pin marking your own location, labelled "My QTH" on click. It reads from the same location (latitude/longitude) set in the HF Utility tab's location bar; if none has been set, it falls back to a default position rather than being omitted.

FT8/WSPR decoder panel – FT8 INTERNAL source, live decodes and propagation map

WSPR Beacons (separate tab from FT8/WSPR):

Despite the shared "FT8/WSPR" naming in the Decoders menu, WSPR beacon decoding is its own tab (**WSPR Beacons** in the COMPACT category), independent of the FT8 SOURCE toggle described above:

- Uses the wsprd command-line decoder; a status bar shows which wsprd binary was found, the current dial frequency, and a countdown to the next decode cycle
- **Two-minute decode cycle**, with a capture indicator (idle/capturing) alongside the countdown
- **Band buttons** for every WSPR allocation from 630m to 10m (630m, 160m, 80m, 60m, 40m, 30m, 20m, 17m, 15m, 12m, 10m), each pre-set to the correct WSPR dial frequency; an **auto-tune SDR to dial freq** checkbox controls whether picking a band also retunes the radio
- ▶️ Start / ■ Stop / ⌫ Clear controls, plus running cycle and spot counters in the tab header
- Spot table plus a **propagation map**: decoded stations plotted with great-circle arcs
- SNR colour coding: green >−10 dB, amber >−20 dB, blue otherwise
- A ↗️ **WSPRnet** link appears once spots exist, opening your session's spots on wsprnet.org

6.6a PSK Reporter Spot Upload (added in w031)

NEXUS can report decoded stations to **PSK Reporter** (pskreporter.info), the community spot-aggregation service used by WSJT-X, JS8Call, fldigi, and other digital-mode software. This is separate from – and does not require – WSJT-X; NEXUS uploads directly from its own decoders.

- **Enable it:** set your callsign in the HF Utility tab's location & callsign bar (Section 4) – required, spots are not sent until a callsign is set – then turn on PSK Reporter uploading there. Your grid locator is derived automatically from the location already set in that same bar; there's no separate locator field to fill in.
- **What gets reported:** decodes from FT8/FT4 on either FT8 SOURCE (the WSJT-X bridge reports via its own backend-side ALL.txt tailing; FT8 INTERNAL forwards each decode's callsign/frequency/mode/SNR from the browser as it decodes), WSPR Beacons, and the CW decoder's Skimmer mode. Each spot includes the decoded station's callsign, frequency, mode, and SNR where available. (FT8 INTERNAL reporting was added slightly later in the w031 cycle than the other three paths – if your build predates that, only WSJT-X-bridge FT8, WSPR, and CW Skimmer will report.)
- **How it works:** spots are sent over UDP directly to PSK Reporter's collector (report.pskreporter.info:4739) using the same simplified IPFIX protocol WSJT-X and fldigi use – there's no NEXUS-hosted server in between and nothing else to configure.
- **Rate limiting:** to be a good citizen of the shared service, uploads are batched and rate-limited (at most roughly one report burst every few minutes, following PSK Reporter's own published guidance) rather than sending a datagram per individual decode.
- **Callsign extraction:** for FT8/WSPR this is straightforward – the decoder already reports the sending station's callsign directly. For CW Skimmer spots, NEXUS applies the same CQ/DE heuristic parsing used to populate the Skimmer's own callsign column; if a message doesn't clearly contain a recognisable callsign, it's simply not reported rather than guessed at.
- **Verifying it's working:** once enabled, your own spots should appear on pskreporter.info within a few minutes of a decode – search by your callsign on their live map/query page.

6.7 ADS-B

Requires dump1090-fa or readsb running locally. NEXUS polls the JSON feed at http://127.0.0.1:8080/data/aircraft.json (falling back to :8080/skyaware/..., then :8754/data/aircraft.json, then the SBS text feed on port 30003).

- Aircraft table: callsign, squawk, altitude, speed, heading, distance
- Live map (Leaflet) with aircraft icons
- Click aircraft to tune to its frequency and show details

6.8 ACARS

Requires multimon-ng or acarsdec. Needs **Full IQ** mode (see section 6 above).

- Message table with flight, registration, label, content
- Detail panel: full message text
- Filter by flight or registration

6.9 AIS

NEXUS now includes a **native AIS decoder (added in w0.2.6)** – no external program required. It reads the same Full IQ stream already used by the other Full IQ decoders (WEFAX, ACARS, POCSAG), so there's nothing extra to install or run:

- Tune to **CH87B (161.975 MHz)** or **CH88B (162.025 MHz)** – the Quick Tune buttons below – and confirm SDRConnect's stream mode is **Full IQ** (see section 6 above).
- Turn on the existing **AIS Start/Stop** toggle, in either the AIS tab or the Marine tab. There is no separate "native" switch – it's the same control already used for the UDP method below; NEXUS decodes locally whenever Full IQ is active and tuned near an AIS channel.
- Internally it demodulates GMSK at 9600 baud directly off the IQ stream: FM-discriminates, filters, recovers the bit clock via a PLL, locates HDLC frame boundaries, removes bit-stuffing, and checks CRC-16. Only CRC-verified frames are ever shown, so there are no false vessel entries.
- Verified against real AIS captures, correctly recovering position reports and other AIS message types (binary broadcast, safety messages, UTC inquiries), with the same vessel MMSIs independently confirmed across separate recordings taken minutes apart.

The older AIS-catcher / UDP method still works exactly as before and is not being replaced – it's a separate external program (AIS-catcher) that decodes AIS itself and sends NMEA sentences to NEXUS over UDP port 10110. This is still useful if you want to share one AIS-catcher feed across multiple programs at once, or already have it set up from before w0.2.6. Both the native decoder and the UDP feed write into the same vessel table, so running both simultaneously just means more complete coverage – they don't conflict.

- Vessel table: MMSI, name, type, speed, heading, distance
- Map view with vessel positions
- Click vessel for details
- Quick Tune buttons: CH16 (156.800 MHz, distress/calling), CH87B (161.975 MHz), CH88B (162.025 MHz) – the two data channels AIS actually broadcasts on, plus the voice/calling channel

Fixed in w0.2.4: speed, course, and heading previously always showed as "–" and every vessel icon rendered with the same default heading, regardless of the ship's actual heading. The backend decoder was emitting these fields under different internal names (sog/cog/hdg) than the ones the table/map/popup actually read (speed/course/heading), so the values never reached the display. Navigational status and destination – both with dedicated UI – were also never being extracted from the AIS payload. All four fields now populate correctly.

Marine tab no longer embeds AIS (w032): earlier builds duplicated the CH16/CH87B/CH88B Quick Tune buttons and a vessel list on the Marine/VHF tab. That tab is audio-mode only (works on any stream mode, no decoder of its own), so an AIS panel – which needs Full IQ – didn't belong there. It's been removed; the Marine/VHF tab now just tunes voice channels, with an **Open AIS tab** button that jumps to this dedicated AIS tab for vessel tracking.

Native GMSK decoder (added in w0.2.6): see above – NEXUS decodes AIS locally off the Full IQ stream without requiring AIS-catcher or any external program. The AIS-catcher/UDP path described above remains available as an alternative for shared-feed setups.

Silent no-op in Compact/IQ-Lite mode fixed (w031): turning on AIS while SDRConnect's stream mode was Compact or IQ-Lite used to do nothing at all, with no on-screen indication of why – every other Full-IQ-only decoder already showed an IQ-mode warning toast in this situation, but the AIS start handler was the one exception that skipped this check. AIS now shows the same warning as WEFAX/ACARS/POCSAG when started without a genuine Full IQ stream.

Decode accuracy fixed (w032): the native decoder's bit-slicer compared the raw FM-discriminator output against a fixed zero threshold, with no compensation for the small carrier offset every real-world signal has (AIS transmitters are allowed up to ±500 Hz of tolerance, on top of whatever error the receiving SDR's own local oscillator has). That was enough to corrupt bit sync before a single CRC-verified frame could ever be produced, even with genuine AIS signal clearly visible on the waterfall. A DC/carrier-offset tracker was added to the decode chain to correct for this.

Front-end DSP overhaul (w032): cross-checking the native decoder against AIS-catcher (an external reference decoder) on the same recorded signal showed NEXUS recovering noticeably fewer messages. Investigation traced this to the decoder trying to bit-sync continuously across background noise rather than only during real transmissions, and to a receive filter that wasn't shaped to AIS's actual GMSK modulation – not, as first suspected, the bit-sync clock recovery itself. The decode chain now detects real burst energy before attempting to decode at all, and filters with a receive filter matched to AIS's modulation rather than a generic one. Measured on a real recorded capture: roughly double the unique messages and vessels recovered versus the previous release, with no loss of previously-working decodes. Messages that carry no navigational or name data (types 8, 10, 12 – see the Troubleshooting guide) can still dominate depending on local traffic, and matching a dedicated tool like AIS-catcher fully would need a larger rework (coherent frequency tracking, soft-decision decoding) not included in this pass.

> **w033 – second AIS front end (Dire Wolf-derived), runs alongside the existing one:** this fork adds a second native GMSK/HDLC decoder, ported from the multi-slicer bit-sync approach used by Dire Wolf (an open-source AX.25/APRS packet decoder), running concurrently with the existing native decoder above – not replacing it. Both read the same Full IQ stream, and both feed the same vessel table, merged by MMSI: if either decoder produces a CRC-verified frame for a vessel, it shows up, and if both decode the same vessel, their results merge rather than duplicate. Validated against 3 real captured WAV files: the combined pair matched or beat either decoder running alone on every file. This is w033-only – w032 has only the single native decoder described above.

6.9a Online Vessel Name Lookup (aisstream.io)

Not every vessel broadcasts its own name over the air – name/callsign only come from AIS message types 5 and 24 ("static and voyage data"), which transmit far less often than position reports, so a vessel can sit in the table showing only its MMSI for a while even with a perfectly good decode. NEXUS can optionally cross-reference MMSIs it has already heard (even via message types with no name/position of their own, e.g. 8/10/12) against aisstream.io – a free (beta) global WebSocket AIS feed aggregated from thousands of contributor receivers worldwide – to pull in a fuller decode of the same vessel from another receiver.

> **You need your own key – NEXUS never ships with one built in.** Sign in at aisstream.io/authenticate (free, GitHub or other supported login) to generate one, then paste it into the **aisstream.io feed** field in the AIS tab and click **Save** – no restart needed.

- The AIS tab shows the current status next to the input field: **not configured**, or **configured (…XXXX)** showing the last 4 characters of whatever key is active, so you can confirm it saved without displaying the whole key on screen.
- Unlike a capped per-call lookup, this is one persistent subscription: NEXUS periodically resubscribes with whichever tracked MMSIs still lack a name or position (up to 50 at a time, aisstream.io's own limit), worldwide, so it isn't tied to your configured location and works the same wherever NEXUS is running.
- Resolved fixed/durable fields (name, callsign, ship type, destination) persist to disk across restarts, so a vessel resolved once is never re-queried – dynamic fields like position and speed are deliberately never persisted for moving vessels, since they go stale within minutes and shouldn't be replayed later as if current.
- Names filled in this way are tagged with a small 📡 icon in the vessel table, so it's always clear which names came from an actual decoded RF broadcast versus aisstream.io's live feed.
- Clicking **Clear** in the AIS tab removes a saved key entirely.

MMSI plausibility filter (w032): NEXUS rejects any decoded MMSI whose structure can't correspond to a real ITU-assigned identity (e.g. a Maritime Identification Digit outside the 201-775 range assigned to ships, or the AIS Aid-to-Navigation 99MIDxxxx format) before it's ever added to the vessel table or looked up online – this keeps both the displayed list and aisstream.io queries limited to structurally plausible vessels.

If you're building NEXUS from source: the build scripts (build_macOS.sh / build_Windows.bat) deliberately never bundle an aisstream.io key – they check for and warn about `.aisstream_key.json` if one happens to exist in your local copy, and will abort the build if one is ever found packaged inside the output app. This is intentional: every user is expected to generate and enter their own key via the GUI field above, not inherit the developer's.

6.10 HFDL

Requires dumphfdl (JSON output on port 5556).

- Message table with ground station, frequency, aircraft
- Detail panel

6.11 VDL2

Requires dumpvdl2 (JSON output on port 5557).

6.12 POCSAG / Multimon

Requires multimon-ng. Works on every stream mode (Compact, PCM, IQ-Lite, Full IQ) – earlier builds' status bar only updated live on the RTL-SDR-direct path; this is fixed in w0.2.6, see below.

- Mode selector: POCSAG512, POCSAG1200, POCSAG2400, FLEX, AFSK1200/APRS, EAS, DTMF, ZVEI1/2/3, DZVEI, PZVEI, EEA, EIA, CCIR, FSK9600, MORSE_CW – tick any combination and click Apply
- Status bar: live audio level meter with peak-hold, activity pulse dot, decode rate (/min), per-mode decode counters on each active mode pill, PID and uptime
- Message list with capcode and alpha text; new decodes flash briefly as they arrive and the total counter animates on each increment
- Quick Tune panel: known POCSAG (UK) and APRS frequencies, plus FLEX pager (UK/EU and US bands) and NOAA Weather Radio (a common place to catch EAS activations in the US) – see below

Status bar fixed for all stream modes (w0.2.6): the audio-peak/PID/uptime heartbeat used to be broadcast from one specific code path written for RTL-SDR-direct operation. Every other stream mode – Compact, PCM, IQ-Lite, and Full IQ, i.e. the SDRConnect-bridge paths most people actually use – never received it after the initial one-time status at decoder start, so the meter and uptime readout looked frozen even while multimon-ng was decoding correctly in the background. The heartbeat now broadcasts from inside the decoder's own audio-processing method, so it fires on every stream mode uniformly.

"Is it actually running" bling (w0.2.6): the audio meter now has a peak-hold tick that lingers briefly after a level spike instead of snapping back down instantly, plus a small scrolling strip-chart of recent audio levels next to it. The activity dot pulses with a brief expanding ring on each decode. Each active mode pill shows a running decode count. New rows in the message list flash on insert, and the total decoded counter pops/scales on every increment – all purely cosmetic, but intended to make "is this thing doing anything" obvious at a glance rather than requiring you to watch for new rows.

Quick Tune – what has a button and what doesn't: POCSAG, APRS, and FLEX all use fixed, well-known frequency blocks (UK/EU pager and APRS allocations, plus separate UK/EU and US FLEX bands), so Quick Tune buttons jump straight to them. EAS has no frequency of its own – it rides on whatever broadcast audio you're tuned to – so the EAS buttons instead tune to NOAA Weather Radio (162.400/162.550 MHz, US), one of the most common real-world sources of EAS activations. DTMF and the selcall modes (ZVEI1/2/3, DZVEI, PZVEI, EEA, EIA, CCIR) deliberately have no Quick Tune buttons: DTMF can appear on any voice channel, and selcall channels are assigned per country/agency with no universal frequency to offer. MORSE_CW likewise has no button here since it can be decoded from anywhere in the HF CW bands – use the dedicated CW tab for that.

6.13 Digital Voice (DSD+)

Requires DSD / DSD+, reading from a virtual audio input (BlackHole on macOS, VB-Audio Cable on Windows) – NEXUS auto-launches it if found on PATH.

- Supported modes, auto-detected from audio: **DMR** (Tier II/III trunked), **P25** (Phase 1 & 2, FDMA/TDMA), **D-STAR** (ICOM amateur), **C4FM/Fusion** (Yaesu amateur), **NXDN** (Kenwood/Icom commercial), **dPMR** (Tier 1 simplex)
- Decoded rows show mode, ID/talkgroup, and detail columns, plus a live text/info stream below
- This decoder plays back voice frames it decodes; it does not track a trunked system's control channel across calls – for that, see Section 6.18, Trunked P25/DMR/NXDN

6.14 Numbers-Station / HF-Intel (Rivet)

No external tool required – a clean-room NEXUS-native decoder (built from public ITA2/DSC specifications, not from any third-party source).

- Mode toggle: **DSC** (Digital Selective Calling, 100 baud FSK) or **Baudot** (ITA2 5-bit teleprinter code, classic numbers-station format)
- Quick Tune buttons jump straight to known DSC/Baudot centre frequencies and set the matching baud rate and shift
- Live decoded text/message panel, updated as symbols are demodulated
- Status badge shows active/inactive and current mode
- Start/Stop control independent of other decoders (starting Rivet deactivates any other active decoder, matching the single-decoder-at-a-time COMPACT behaviour)

6.15 FreeDV HF Digital Voice

Requires freedv_rx (part of the codec2 project) on your PATH. See the Quick Start guide for build instructions.

- Mode dropdown: 1600, 700C, 700D, 700E, 2400A, 2400B, 800XA – pick the mode matching the signal you're tuned to
- Genuine real-time audio playback through your browser speakers (Web Audio pipeline, not a file save) – gapless scheduling so decoded speech plays back smoothly
- Mute button and volume slider for the decoded audio
- Live VU meter showing decoded audio level
- Sync/SNR status badge – shows whether freedv_rx has achieved frame sync on the signal, and the current SNR estimate
- Start/Stop control; switching modes while running automatically restarts the decoder with the new mode
- NEXUS sets a 2000 Hz demodulation bandwidth automatically when you click Start

6.16 WEFAX

Decodes HF weather-fax images. ○ **FULL IQ** – needs a genuine raw IQ stream, won't work in Compact mode.

- Quick Tune buttons for Northwood (UK) HF WEFAX on 2618.5, 4610, and 8040 kHz – all USB, 2.4 kHz bandwidth
- Decoded images render directly in the tab as they build, line by line
- Start/Stop control, same pattern as every other decoder

6.16a SSTV (Slow-Scan Television)

New in w033 – a native NEXUS decoder, no external binary required. Decodes Martin M1/M2 and Scottie S1/S2/DX. ○ **FULL IQ** – needs a genuine raw IQ stream, won't work in Compact mode.

- Tune to an SSTV transmission (typically 14.230 MHz USB on 20m) and click Start
- The decoder searches for the VIS (Vertical Interval Signalling) leader tone that identifies the mode, then decodes the image line by line as it arrives
- The image builds progressively in the tab as each line completes – no need to wait for the whole transmission to finish before seeing a partial picture
- Robot 36/72 transmissions are detected via their VIS code (so you'll see that a Robot mode signal is present) but are not yet decoded to an image

Note: validated against synthetic test signals fed through the same FM-discriminator/IQ pipeline the real code uses; real-world accuracy against actual on-air signals (noise, Doppler, clock drift) hasn't been separately confirmed yet.

6.16b rtl_433 (ISM-Band Sensors)

New in w033 – decodes short-range ISM-band (typically 433/315/868 MHz) sensor transmissions via the external **rtl_433** tool (auto-launched by NEXUS if found on PATH, same subprocess-plus-UDP-JSON pattern already used for HFDL/VDL2). ▣ **EXTERNAL** – a separate feed/process, not gated by SDRConnect stream mode.

- Live table of readings as they're received: sensor type/model, ID, and whatever fields that sensor type reports (temperature, humidity, tyre pressure, etc.)
- Covers common consumer ISM-band devices: TPMS (tyre-pressure monitors), wireless weather stations, and smart utility meters, among others rtl_433 itself supports
- Requires the rtl_433 binary to be installed and reachable on PATH; NEXUS does not bundle it

6.17 DAB / DAB+

Decodes a full DAB/DAB+ ensemble (multiplex) via dab_radio_nexus, a headless streaming tool NEXUS auto-launches if found on PATH – built on top of williamyang98/DAB-Radio (MIT licensed), replacing the dab-cmdline engine used through w033. ◻ FULL IQ – an ensemble spans roughly 1.5 MHz, so Compact-mode streaming can't capture a full multiplex; Full IQ is required.

New engine, new hardware reach (w035): dab_radio_nexus owns no hardware itself – it reads IQ from its own stdin and writes PCM audio to its own stdout, so it works with any IQ source NEXUS can already reach: a direct RSPdx, a networked nRSP-ST, or an RTL-SDR. The previous dab-cmdline engine opened the SDRplay API itself and only worked with a directly USB-attached RSPdx – it couldn't see a networked nRSP-ST at all.

Band III quick-tune grid with background Scan: click any channel to retune directly, or click Scan to sweep every Band III channel automatically – channels are colour-coded as the currently tuned channel, a genuine ensemble found there, a channel tested with nothing found, or the channel currently being tested.

Every discovered service in a locked ensemble decodes simultaneously, so picking a different station is an instant buffer switch, not a decoder relaunch – only starting/stopping the engine, or changing the DAB channel itself (which retunes the actual hardware), restarts the underlying decoder.

Dedicated player column showing the selected station, with a live Dynamic Label (DLS) scrolling text ticker – song/artist, news, weather, or traffic, entirely up to what that broadcaster sends – and a MOT Slideshow image (station logo or album art) whenever the station transmits one.

The locked ensemble's name and frequency are shown prominently in the top bar, with a status dot that lights green once a real ensemble is locked.

A Technical Details drawer (⚙ icon) shows RF frequency, SNR, decoder state, buffer fill, device, and IQ rate for troubleshooting.

6.18 Trunked P25/DMR/NXDN

Follows a trunked radio system's control channel across calls, via **OP25** or **trunk-recorder** (either auto-launched by NEXUS if found on PATH). ▣ **EXTERNAL** – a separate feed/process, requiring Full IQ (following calls across voice channels needs wideband capture of the whole site).

- Supported systems: **P25** (Phase 1 & 2, control + voice), **DMR** (Tier III trunked – Cap+/Connect+), **NXDN** (Type-C trunked – NXDN96/48)
- Enter the system's control-channel frequency (e.g. 851.0125 MHz) and click **Set Control Channel** to begin tracking
- Live table of system, talkgroup, frequency, and call status as the tracker follows the site
- This is a distinct feature from Section 6.13's Digital Voice (DSD+) decoder – DSD+ decodes voice audio on a single tuned channel; this tracks and follows an entire trunked site's control-channel signalling

6.19 DRM (Digital Radio Mondiale)

Decodes DRM and DRM+ digital shortwave/FM broadcasts via dream_nexus, a headless streaming tool NEXUS auto-launches if found on PATH – built on the open-source Dream DRM receiver engine. ◻ FULL IQ – DRM is an OFDM signal spanning up to ~20 kHz (DRM30) or ~100 kHz (DRM+); Full IQ is required, the same as DAB.

- Auto-set tuning – you don't need to configure SDRConnect's sample rate, mode, or bandwidth by hand for DRM. Whatever sample rate SDRConnect is already streaming at in Full IQ mode (250 kSPS has been tested live against a real off-air station) works; dream_nexus does its own internal resampling and channel filtering from the raw IQ NEXUS feeds it.
- Quick-tune card grid – saved DRM stations appear as individual card buttons, grouped by category, in place of the older scrollable list; click a card to retune directly to it. Cards can be added from whatever frequency is currently tuned and removed individually.
- 3-column tab layout – controls (frequency, robustness mode) on the left, the quick-tune card grid in the centre (the tab's main working area), and lock status plus the now-playing audio player on the right.
- Robustness mode – DRM defines four OFDM robustness modes (A–D) trading resilience against multipath/fading for data capacity; select to match the broadcaster if known, or leave on auto-detect.
- Live audio playback – decoded audio streams to the built-in player as soon as the receiver locks. If playback stutters or sounds slowed/slurred, hard-reload the browser (Cmd+Shift+R / Ctrl+Shift+R) – see the Troubleshooting guide's DRM Issues section for the underlying cause and fix.
- Packaged macOS releases bundle dream_nexus automatically as of w035 (built with a pinned deployment target so it also launches correctly on older macOS versions, not just the build machine's own – see the Troubleshooting guide's “Companion engine won't launch on an older Mac” entry). Building NEXUS from source still requires dream_nexus reachable on PATH; see the DRM/Dream build guide included with this release for build instructions.

6.20 HD Radio (NRSC-5/iBOC)

Decodes HD Radio (NRSC-5) hybrid digital AM/FM broadcasts via nrsc5_nexus, a headless streaming tool NEXUS auto-launches if found on PATH – built on the open-source theori-io/nrsc5 receiver engine (GPLv3). ◻ FULL IQ – HD Radio's digital sidebands extend several hundred kHz around the analog carrier; Full IQ is required, the same as DAB and DRM.

HD Radio (NRSC-5) is a North America-only broadcast standard – there is no HD Radio service in the UK or Europe, so this decoder has nothing to receive outside North America regardless of antenna or SDR setup.

- Auto-set tuning – like DAB and DRM, you don't need to configure SDRConnect's sample rate by hand; whatever rate SDRConnect streams in Full IQ mode, NEXUS's own resampler converts internally to nrsc5's required native rate (confirmed against a real captured off-air NRSC-5 broadcast end-to-end; not yet tested against a live over-the-air signal).
- FM/AM mode – HD Radio broadcasts on both AM and FM; select the mode matching the station you're tuned to before starting the decoder.
- Multiple simultaneous programs (HD1-HD8) – an HD Radio FM signal can carry up to 8 digital subchannels alongside the main analog program; NEXUS decodes and buffers audio for every active program at once, not just the primary one, with per-program station name, slogan, and now-playing ID3 metadata (title/artist/album/genre) shown as it arrives.
- Live audio playback – decoded audio streams to the built-in player per program as soon as the receiver locks.
- Packaged macOS releases bundle nrsc5_nexus automatically as of w035 (built with a pinned deployment target so it also launches correctly on older macOS versions, not just the build machine's own – see the Troubleshooting guide's "Companion engine won't launch on an older Mac" entry). Building NEXUS from source still requires nrsc5_nexus reachable on PATH; see the HD Radio build guide included with this release for build instructions.

7. Airband Tab

Shows nearby airport frequencies based on your location (if provided) or manual entry.

Column layout improved (w031): the three columns now use proportional widths (rather than fixed pixel widths) so they make better use of wide windows, and the UTILITY column's frequency list reflows into a denser multi-column grid instead of a single long list.

- Fetches OurAirports database on first use (auto-downloaded)
- Lists ground, tower, approach, departure, ATIS, CTAF frequencies
- Click any frequency to tune
- AIS vessel count from the AIS decoder is shown in the header

Correcting a wrong airport frequency:

- OurAirports data is occasionally out of date (a tower frequency changes, an ATIS is decommissioned, etc.). Click the ✎ button next to any frequency row to open a small correction box where you can edit the frequency and/or description, or hide the row entirely.
- Corrections are stored locally on your machine (not sent anywhere) and are re-applied automatically every time that airport's data loads, including after the next OurAirports refresh.
- Click Revert in the correction box to discard your edit and go back to the original OurAirports value.
- This only affects what's displayed in NEXUS – it doesn't change the OurAirports database itself.

VOLMET coverage:

- The VOLMET list covers VHF stations across Europe/UK and HF stations worldwide (Americas, Asia/Pacific, Atlantic). This matches how the real VOLMET network operates – VHF VOLMET is essentially a European/UK-only service, while everywhere else en-route aviation weather is broadcast on HF instead, so HF is the only way to get VOLMET coverage outside Europe.

Bookmarking the tuned frequency:

- Click the ⭐ Bookmark button on the top command row to save whatever frequency you're currently tuned to. A popup opens prefilled with the best available match from your Frequency Lists, cached Nearby Airports data, or the SigID database – edit the name, notes, or mode if needed, then click Save Bookmark. If nothing matches, the fields are left blank for you to fill in manually.

FM Nearby panel and offline fallback:

- The FM Nearby panel looks up local FM broadcast stations for your current location via a live internet lookup (radio-browser.info), shown as "(live lookup)" in the panel header.
- If that lookup doesn't respond within 7 seconds (no internet, or the lookup service is unreachable), the panel automatically falls back to FM-band entries (87.5–108 MHz) from your own imported Frequency Lists – see Section 9, Frequency Lists Tab – shown as "(from your imported list)" in amber. Entries from a region-matching list are preferred; if none match, any FM-range entry is shown instead.
- If live data arrives late, it replaces the fallback data automatically.
- If neither the live lookup nor any imported list has usable FM entries, the panel explains this directly and points you to the 📂 FREQ LISTS tab to import a list with FM-band entries.

AIRBAND tab – nearby airports, VOLMET, and utility/guard/data frequencies

8. Bookmarks (BM Tab)

Right-click on either the **spectrum** or the **waterfall** to bookmark the frequency under the cursor – you'll be prompted for a name, then the bookmark is saved. Bookmarks appear as clickable markers/labels on the spectrum and waterfall.

You can also bookmark the currently-tuned frequency from anywhere using the ⭐ Bookmark button on the top command row (see Section 7, Airband Tab, above). A few result lists (Signal Intelligence) still have their own bookmark shortcut alongside each entry.

Bookmark popup keyboard behaviour (w0.2.6): inside the popup, pressing **Enter** in any field saves the bookmark immediately (equivalent to clicking Save Bookmark), and pressing **Escape** cancels and closes the popup. Typing in the Name, Notes, or Mode fields no longer accidentally triggers NEXUS keyboard shortcuts (r, c, d, p, etc.) – earlier builds fired those shortcuts on every keypress while an input field was focused, which could, for example, toggle Cinematic mode when the user typed 'c', making the popup appear to vanish.

Click any bookmark label on the spectrum to tune straight to it – label hit-detection uses the exact drawn geometry of the label pill, so clicks register reliably even on tightly-packed labels.

BM tab shows all bookmarks with name, frequency, mode, and notes. Click any bookmark to tune. Delete button removes it.

Bookmarks are saved to darksky_bookmarks.json in the application folder.

9. Frequency Lists Tab

The **FREQ LISTS** tab holds your imported and manually-entered frequency lists, separate from the built-in EIBI/SigID database (though entries from here still surface in SIG INTEL / SW SCHEDULE lookups, tagged **CUSTOM**).

Importing:

- Drag a CSV, TXT, or XLSX file onto the tab, or use its file picker.
- The parser auto-detects columns by header text (case-insensitive, any order): frequency (header containing "freq", "khz", or "mhz"), station/name, mode, language, time/UTC, and site/transmitter.
- Frequency units are inferred from header text and value magnitude.
- XLSX import requires the SheetJS library to be available in-browser; if it isn't, you'll be prompted to save as CSV/TXT instead.
- On import you're prompted for a **list name** and a **region tag** – what the list is for and where it applies (e.g. "Colorado", "UK", or "Global" for worldwide lists). Leave it blank for Global.

Table and tuning:

- All entries across all lists appear in one searchable, filterable table.
- Click any row to tune to that frequency.
- Each list has a visibility toggle (show/hide its entries), a colour tag, and a delete option.
- Add, edit, or delete entries manually – including in the always-present "My Frequencies" list – and export any list back to CSV.

Filters (added in w0.2.6):

- **Near VFO** – when toggled on, the table shows only entries within 5 kHz of your currently tuned frequency. Updates automatically as you tune, so you can pan across a band and see matching schedule entries in real time.
- **On Now** – filters to entries whose broadcast window covers the current UTC time (based on the time/UTC or On/Off columns in your imported file). Lights up when active. If all entries still show after toggling it on, your list's time column wasn't detected – see the Troubleshooting guide's "On Now filter does nothing" entry.
- **All Languages** dropdown – filters to a single language from your imported data. Populated from the Language column across all loaded lists; entries with no language data don't appear in the dropdown.
- **All Sources** dropdown – filters to a single source (list name) – equivalent to the per-list visibility toggle but quicker to reach from the filter bar.

SwSKEDS format support (w0.2.6): Lists exported from Roger Need's SwSKEDS shortwave schedule tool are correctly recognised – frequency, station name, language, and transmitter site columns are all parsed. Broadcast times in SwSKEDS use 'On' and 'Off' column headers (e.g. 0700 / 1900); earlier builds didn't detect these columns, so the On Now filter had no time data to filter against. As of w0.2.6 these columns are parsed and combined into a HHMM-HHMM time string per entry. If you imported a SwSKEDS list before updating, delete and re-import it – existing entries already in localStorage have empty time fields that can't be retroactively filled.

Region tagging and location awareness:

- Each list's region tag is compared against your current location and shown as a colour badge next to the list name: green "In range"/"Global" if it matches your current location, amber with the tag name if it doesn't.
- This is informational only – lists are never hidden or filtered out based on region, just flagged, so out-of-region lists stay fully usable.
- Click a list's region badge to edit its tag at any time.
- Badges refresh automatically whenever you change your location.

Storage:

- Imported and manually-entered lists are stored in browser `localStorage` (key `hf_custom_lists`) and persist across restarts.
- Internally, `hfCustomListLookup(freqMhz, tolMhz)` searches all loaded lists for entries within a tolerance of the tuned frequency (default ±5 kHz, matching the built-in HF lookup tolerance).

9a. Reporting / Logbook Tab

New in w033 – the 📒 **REPORTING** tab is a general-purpose logbook for anything you receive: SWL reception reports, FT8 decodes, and any other signal worth recording. It isn't limited to amateur-radio QSOs – log a broadcast station, an airband transmission, an AIS contact, or a numbers station just as easily as an FT8 exchange.

Log table:

- Columns: time (UTC), callsign, frequency (MHz), mode, SNR, grid square, mission tag, and notes.
- Click any column header to sort by it; click again to reverse the sort direction.
- Results are paginated (40 entries per page) with Prev/Next controls at the bottom of the table.

Search and filters:

- The search box matches against callsign, grid, notes, and mission tag.
- Dropdown filters for **Mode**, **Band** (derived automatically from each entry's frequency), and **Date** narrow the table further; all filters combine (AND, not OR).
- **Mission tag chips** – click a chip (FT8, HF Utility, Airband, Marine VHF, Broadcast, AIS, SIGINT, General SWL, or any custom tag you've used) to filter to just that tag; click it again to clear the filter. Chips are colour-coded and the same colour appears in the table's Mission column and export output.
- **✕ Clear** resets search, all dropdown filters, and the active mission chip in one click.

Adding and editing entries:

- Click **+ New Entry**, or press **Ctrl+N** from anywhere in NEXUS, to open the Add Log Entry modal. The frequency field is pre-filled with your currently tuned VFO frequency.
- Click the ✎ icon on any row to edit that entry, or the ✕ icon to delete it (a confirmation step shows the entry's callsign, mode/frequency, and timestamp before it's removed).
- Only a callsign or some notes text is required to save – every other field (frequency, mode, SNR, grid, mission) is optional, so you can log a signal even when you don't have a full ID.
- Selecting **+ Custom…** in the Mission dropdown lets you type your own tag instead of picking from the built-in list.

Statistics panel:

- Click 📊 **Stats** to reveal a row of at-a-glance breakdowns: spots per band, SNR distribution, DX distance (derived from each entry's notes field when a grid-based distance was computed), and spots per mission tag – all rendered as simple bar/histogram rows, no chart library required.
- The stats panel reflects whatever search/filter/chip selection is currently active in the table above it.

FT8 auto-populate:

- Click ▯ **Populate from FT8** (also available from the Reporting tab toolbar) to pull FT8 decodes captured earlier in the session into the logbook automatically.
- Each imported entry gets: timestamp, callsign, frequency, mode (FT8), SNR, grid, mission (tagged "FT8"), and a notes string containing the decoded country, band, distance and bearing from your station (when the sender's grid was decodable), the FT8 time offset (dt), and the raw decoded message.
- Already-logged decodes are skipped automatically – clicking Populate repeatedly, or across multiple FT8 sessions, never creates duplicate rows for the same decode.
- A confirmation toast reports how many new entries were added.

Export:

- ▯ **ADIF** exports the currently filtered/visible rows as a standard ADIF (.adi) log file, suitable for import into most logging software.
- ▯ **CSV** exports the same filtered rows as a plain CSV file with one column per LogEntry field.

Storage:

- Log entries are saved server-side to darksky_logbook.json in the application folder – the same persistence pattern used for bookmarks – so your logbook survives restarts and isn't tied to browser storage.

10. Scan Tab

The ⟳ SCAN tab runs an automated band scanner that steps the VFO across a frequency range looking for active signals. It was originally a small collapsible panel in the bottom-right corner; as of this build it has its own dedicated tab for better visibility, with no change to its underlying behaviour. The spectrum and waterfall remain visible above the tab area while scanning, exactly as with any other tab.

Controls:

- **From / To** – scan range in MHz
- **Step** – frequency increment per scan step (kHz)
- **Dwell** – time spent listening at each step before moving on (ms)
- **SNR threshold** – minimum signal-to-noise ratio (dB) to count as a "hit"
- **Mode** – demodulation mode to use while scanning and on any hit

Running a scan:

- **Start** begins stepping through the range, tuning the VFO and measuring SNR at each step against the live spectrum data.
- **Pause** halts stepping in place (useful to listen longer on a hit) without resetting the scan.
- **Stop** ends the scan entirely.
- Status text shows current frequency and scan state.

Signal hits:

- Whenever a step's SNR exceeds the threshold, the scanner logs a hit (frequency + SNR) to the hit list and pauses there automatically so you can listen.
- Click any hit in the log to tune straight to it.

11. SSH Launcher

If SDRConnect runs on a remote Linux server (e.g. a Raspberry Pi or headless machine), the SSH Launcher starts it automatically over SSH.

Access: click **Command** → SSH Setup, or the Connection Wizard appears on first launch.

Configuration fields:

- **Host** – IP address or hostname of remote server
- **Port** – SSH port (default 22)
- **Username** – SSH login name
- **Password** – SSH password (session-only, never saved to disk)
- **SDRConnect command** – command to start SDRConnect on the remote (default, w035: `./SDRconnect_headless --websocket_port=5454`)
- **Device preference** – Auto / Force USB RSP / Force nRSP-ST (w035). Controls which SDRplay device NEXUS connects to at startup when SDRConnect can see more than one – Auto prefers a directly-connected USB RSP if present, otherwise falls back to Default Device; Force USB RSP never falls back to a networked device; Force nRSP-ST always picks the networked nRSP-ST (or Default Device) even if a USB RSP is also plugged in. Set in the nRSP-ST/Local tab of the Connection Setup wizard.
- **Startup delay** – seconds to wait for SDRConnect to initialise before connecting
- **Device release wait** – seconds to wait after stopping before allowing restart (prevents SIGSEGV in swig_bindings)

SDRconnect Headless (added in SDRconnect 1.0.7) needs no local SDRconnect GUI running at all – it serves the WebSocket API directly from the remote box. If you'd rather run the older SDRconnect --server plus a local SDRconnect GUI, set SDRConnect command to ./SDRconnect --server and set a Local Client path in the wizard; this older mode still works unchanged, it's just no longer the default.

SSH config (minus password) is saved to `~/.darksky_nexus/ssh_config.json`.

The SSH strip in the top bar shows connection status and a Stop button while a session is active.

12. RTL-SDR Mode

When started with --rtl HOST, NEXUS bypasses SDRConnect and connects directly to rtl_tcp.

Start rtl_tcp first:

```
rtl_tcp -a 127.0.0.1 -s 2048000
```

Then start NEXUS:

```
python3 w035_NEXUS.py --rtl 127.0.0.1
```

In RTL-SDR mode:

- Gain control via NEXUS gain slider (LNA index 0–28 for R820T, or AGC)
- Overload-reduce button applies −2 gain steps automatically
- Audio demodulation runs in a separate process (avoids GIL)
- scipy required for best audio quality; fallback to basic FM demod without it
- Volume control active

Supported modes: WFM, NFM, AM, USB, LSB, CW (demodulated in Python)

13. Settings & Persistence

The following are automatically saved and restored:

Setting	Storage
Waterfall height	localStorage
UI scale (80%-160%)	localStorage
Decoder engine (NEXUS/FLDIGI) per tab	localStorage
Bookmarks	darksky_bookmarks.json
SSH config (no password)	~/.darksky_nexus/ssh_config.json
EIBI / airport data	eibi.csv, airports.csv, airport-frequencies.csv

There is no manual Save button – everything persists automatically.

14. Keyboard & Mouse Reference

Mouse

Action	Result
Left-click waterfall	Tune VFO to frequency
Right-click waterfall or spectrum	Save bookmark at clicked frequency
Click bookmark label on spectrum	Tune to that bookmark
Scroll on waterfall	Adjust speed
Shift+Scroll on waterfall	Adjust noise floor
Ctrl+Scroll on waterfall	Adjust zoom
Drag frequency axis	Pan waterfall
Drag resize handle	Resize waterfall height
Click signal label	Look up in EIBI/SigDB
Drag M/S markers (RTTY scope)	Set mark/space tone frequencies
Click RTTY scope canvas	Snap nearest marker to clicked Hz
Drag fldigi waterfall carrier line	Retune fldigi carrier
Click fldigi waterfall	Set carrier to clicked position
Click a DECODERS category tab	Switch to that category's decoder grid

Keyboard

Key	Result
Enter (in VFO field)	Tune to typed frequency
Enter (in Bookmark popup field)	Save bookmark
Escape (in Bookmark popup field)	Close popup without saving
Ctrl+N	Open New Log Entry modal (Reporting tab), from anywhere in NEXUS
Enter (in Log Entry modal field)	Save log entry
Escape (in Log Entry modal field)	Close modal without saving
Escape (global)	Close Bookmark popup / Log Entry modal / Radar / DSP modal / Bands panel
r	Open Radar modal (suppressed while typing in any input)
d	Open DSP modal (suppressed while typing in any input)
c	Toggle Cinematic mode (suppressed while typing in any input)
b	Toggle Bands panel (suppressed while typing in any input)
F5	Reload page
Cmd+Shift+R (macOS)	Hard reload (clears cache)
Ctrl+Shift+R (Windows/Linux)	Hard reload

Note (w0.2.6): Single-letter keyboard shortcuts are suppressed while any text input, textarea, or select element has focus – including inside the Bookmark popup and the Command palette. This prevents typing in a field from accidentally triggering UI shortcuts. As of w033, the same guard covers the Reporting tab's New/Edit Log Entry and Delete confirmation modals.

Appendix A – fldigi Setup for NEXUS

To use FLDIGI engine mode:

1. Install fldigi from https://www.w1hkj.com
2. Install a virtual audio device:
   - macOS: BlackHole (https://existential.audio/blackhole) – 2ch version
   - Windows: VB-Audio Cable (https://vb-audio.com/Cable)
3. In SDRConnect, set audio output to the virtual device (e.g. BlackHole 2ch)
4. In fldigi → Configure → Audio → Capture: set to the same virtual device
5. In fldigi → Configure → Rig Control: set to Hamlib NET rigctl, localhost:4532 (optional – lets fldigi follow NEXUS frequency automatically)
6. Start fldigi. It runs hidden – NEXUS is the interface.
7. In NEXUS, select a fldigi decoder tab, set engine to FLDIGI, click Start.

The fldigi waterfall and decoded text appear in the NEXUS decoder panel.

Appendix B – Frequency Reference

Frequency	Signal
518 kHz	NAVTEX international (English)
490 kHz	NAVTEX national
4209.5 kHz	NAVTEX HF
14.074 MHz	FT8 20m
7.074 MHz	FT8 40m
3.573 MHz	FT8 80m
14.070 MHz	PSK31 20m calling
7.035 MHz	PSK31 40m calling
14.0956 MHz	WSPR 20m
7.0386 MHz	WSPR 40m
1090 MHz	ADS-B Mode S
121.5 MHz	Aviation emergency
156.8 MHz	VHF Marine Channel 16 (distress)

Appendix C – Credits & Acknowledgments

DARKSKY NEXUS is built on top of, and would not exist without, the hardware, software, and open data sources below.

Development

- **Jon Nicol** – creator and developer of DARKSKY NEXUS
- **Claude (Anthropic)** – AI development collaborator: pair-programmed the majority of NEXUS's features across every build in this series, including the native FT8/FT4 decoder integration, the decoder suite, the propagation mapping, and this documentation set

Hardware & Core Platform

- **SDRplay** – NEXUS's primary supported platform. Built for and tested against SDRplay's RSPdx and nRSP-ST receivers and their SDRConnect server software, which NEXUS connects to over WebSocket for spectrum, waterfall, and audio/IQ streaming.
- **RTL-SDR** – USB dongle support via rtl_tcp, for users without SDRplay hardware.

Decoder Engines & External Tools

- **WSJT-X** (Joe Taylor K1JT and the WSJT-X development team) – the FT8/FT4/WSPR bridge source (Section 6.6) reads WSJT-X's own ALL.TXT decode log and sends it frequency-control commands over UDP.
- **wsprd** (part of the WSJT-X project) – the command-line decoder behind the WSPR Beacons tab (Section 6.6).
- **ft8ts** (© Roger Need, GPL v3, https://github.com/e04/ft8ts) – a from-scratch JavaScript/TypeScript port of the WSJT-X FT8/FT4 algorithm, vendored in-browser to power the FT8 INTERNAL native decoder (Section 6.6), requiring no separate WSJT-X install.
- **fldigi** (W1HKJ, https://www.w1hkj.com) – the FLDIGI engine option for CW, RTTY, PSK31, NAVTEX, Olivia, Contestia, MFSK, Hell, and DominoEX (Sections 6.2–6.5, Appendix A).
- **dump1090-fa / readsb** – ADS-B aircraft decode (Section 6.7).
- **dumpvdl2** (szpajder, https://github.com/szpajder/dumpvdl2) – VDL2 aircraft data link decode (Section 6.11).
- **dumphfdl** – HFDL aircraft data link decode (Section 6.10).
- **DSD / DSD+** (szechyjs/dsd; dsdplus.com) – digital voice decode for DMR, P25, D-STAR, C4FM/Fusion, NXDN, and dPMR (Section 6.13).
- **DAB-Radio (williamyang98, https://github.com/williamyang98/DAB-Radio, MIT licensed) –** DAB/DAB+ ensemble decode via dab_radio_nexus, a headless streaming tool built on top of it (Section 6.17), w035 only.
- **OP25** (boatbod) and **trunk-recorder** (robotastic) – trunked P25/DMR/NXDN control-channel tracking (Section 6.18).
- **codec2 / freedv_rx** – FreeDV HF digital voice decode (Section 6.15).
- **multimon-ng** – the Multimon-family decode path (Section 6.12) and related pager/data modes.
- **rtl_433** (Benjamin Larsson and contributors, https://github.com/merbanan/rtl_433) – ISM-band sensor decode (Section 6.16b), w033 only.
- **Dire Wolf** (John Langner WB2OSZ, https://github.com/wb2osz/direwolf) – the multi-slicer bit-sync approach ported to NEXUS's second AIS front end (Section 6.9), w033 only.

Mapping & Visualization

- **MapLibre GL JS** – the FT8 INTERNAL propagation map (Section 6.6).
- **Leaflet** – the WSJT-X bridge and ADS-B maps (Sections 6.6, 6.7).
- **CARTO, OpenStreetMap, and Esri** – the free, no-API-key map tile providers offered in every map style picker.

Data Sources

- **NOAA Space Weather Prediction Center** – solar flux index, K-index, X-ray flux, and band-condition data (Section 4).
- **EIBI** (eibispace.de) and **AOKI** – shortwave broadcast schedule databases (Section 4, Section 5).
- **SigIDWiki** (sigidwiki.com) – the community-maintained signal identification database (Section 4, Section 5).
- **OurAirports** – the airport and frequency database behind the Airband tab (Section 7).
- **radio-browser.info** – live FM broadcast station lookup (Section 4).
- **NCDXF** – the beacon schedule shown in Section 4.
- The amateur radio **DX cluster network** – live DX spots shown in Section 4.
- **PSK Reporter** (pskreporter.info) – the community spot-aggregation service NEXUS can upload FT8/WSPR/CW decodes to (Section 6.6a, added w031).

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